

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

Mohammed Mahmoud Elnaggar

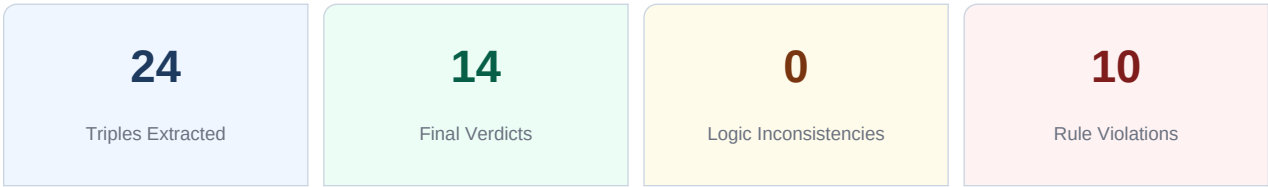
Generated on: 2026-04-26 11:07 UTC

REPORT OVERVIEW & METHODOLOGY

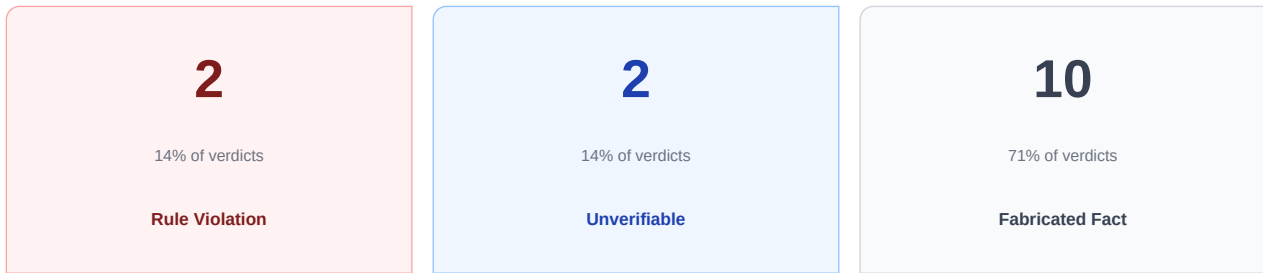
This document outlines the results of the KG-TripleGuard verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	890	HAS_INTEREST_RATE	30		The interest rate associated with the entity.
2	290	HAS_INTEREST_RATE	20		The interest rate associated with the entity.
3	333	HAS_INTEREST_RATE	40		The interest rate associated with the entity.
4	333	HAS_RISK_TIER	high prime		The risk category or tier assigned to the entity.
5	598	HAS_RISK_TIER	high prime		The risk category or tier assigned to the entity.
6	633	HAS_RISK_TIER	high prime		The risk category or tier assigned to the entity.
7	729	HAS_RISK_TIER	high prime		The risk category or tier assigned to the entity.
8	833	HAS_RISK_TIER	high prime		The risk category or tier assigned to the entity.
9	890	HAS_RISK_TIER	high prime		The risk category or tier assigned to the entity.
10	290	HAS_RISK_TIER	high prime		The risk category or tier assigned to the entity.
11	301	HAS_RISK_TIER	deep subprime		The risk category or tier assigned to the entity.
12	333	HAS_APPROVAL_ODDS	200		The estimated odds or likelihood of approval for the entity.
13	290	HAS_APPROVAL_ODDS	30		The estimated odds or likelihood of approval for the entity.
14	890	HAS_APPROVAL_ODDS	40		The estimated odds or likelihood of approval for the entity.
15	300	HAS_INTEREST_RATE	2		The interest rate associated with the entity.
16	303	HAS_INTEREST_RATE	3		The interest rate associated with the entity.
17	333	HAS_INTEREST_RATE	4		The interest rate associated with the entity.
18	777	HAS_INTEREST_RATE	6		The interest rate associated with the entity.
19	loan_100000_usd	HAS_REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require a certain debt-to-income ratio, collateral, or savings (e.g., 10-20% of the loan amount), but this depends on the lender's policy.		The required account balance or financial status needed for the loan.
20	300	HAS_DESCRIPTION	Deep subprime (very poor credit score, indicating high risk of default and limited access to credit)		The textual description or definition of the entity.
21	300	HAS_APPROVAL_ODDS	1		The estimated odds or likelihood of approval for the entity.
22	305	HAS_APPROVAL_ODDS	2		The estimated odds or likelihood of approval for the entity.
23	405	HAS_APPROVAL_ODDS	3		The estimated odds or likelihood of approval for the entity.
24	730	HAS_APPROVAL_ODDS	4		The estimated odds or likelihood of approval for the entity.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
890	HAS_INTEREST_RATE	30	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
290	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	40	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
598	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
633	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
729	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
833	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
890	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
290	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
301	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
333	HAS_APPROVAL_ODDS	200	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
290	HAS_APPROVAL_ODDS	30	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
890	HAS_APPROVAL_ODDS	40	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
300	HAS_INTEREST_RATE	2	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
303	HAS_INTEREST_RATE	3	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	4	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
777	HAS_INTEREST_RATE	6	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM

Subject	Original Predicate	Object	Canonical Predicate	Conf.
loan_100000_usd	HAS_REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require a certain debt-to-income ratio, collateral, or savings (e.g., 10-20% of the loan amount), but this depends on the lender's policy.	HAS_REQUIRED_ACCOUNT_BALANCE	LOW
300	HAS_DESCRIPTION	Deep subprime (very poor credit score, indicating high risk of default and limited access to credit)	HAS_DESCRIPTION	LOW
300	HAS_APPROVAL_ODDS	1	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
305	HAS_APPROVAL_ODDS	2	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
405	HAS_APPROVAL_ODDS	3	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
730	HAS_APPROVAL_ODDS	4	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM

FACTUALLY INCONSISTENT TRIPLES

All triples labelled `factually_inconsistent` from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	30	<code>c_interest_rate_range</code>	The subject credit score 890 is greater than 850, which violates the constraint.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	290	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	<code>c_interest_rate_range</code>	The subject credit score 290 is less than 300, which violates the constraint.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	40	<code>c_interest_rate_range</code>	The object interest rate 40% is greater than 30, which violates the constraint.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	598	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	<code>c_tier_580_619</code>	The subject 598 is in the range 580-619, but the object 'high prime' does not contain 'subprime'.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	633	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	<code>c_tier_620_659</code>	The subject 633 is in the range 620-659, but the object 'high prime' does not contain 'near prime'.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	833	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	<code>c_tier_740_850</code>	The subject 833 is in the range 740-850, but the object 'high prime' does not contain 'super prime'.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	333	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	200	<code>c_approval_odds_range</code>	The object (approval odds %) is 200, which is greater than 100.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	290	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	30	<code>c_approval_odds_range</code>	The subject (credit score) is 290, which is less than 300.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	890	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	40	<code>c_approval_odds_range</code>	The subject (credit score) is 890, which is greater than 850.
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	405	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	3	<code>c_approval_odds_range</code>	Subject (credit score) 405 is outside the valid range of 300-850.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
333	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■ ■■■ 0.00	No DB candidates found for the subject '333' with the given predicate.
729	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■ ■■■ 0.00	No DB candidates found for the subject '729' with the given predicate.
890	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■ ■■■ 0.00	No DB candidates found for the subject '890' with the given predicate.
290	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■ ■■■ 0.00	No DB candidates found for the subject '290' with the given predicate.
301	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	●●●●●■ ■■■ 0.60	The predicate and object match DB candidates, but the subject '301' does not appear in any candidate. Ev: (375 RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL deep subprime); (380 RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL deep subprime); (385 RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL deep subprime); (389 RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL deep subprime); (390 RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL deep subprime); (395 RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL deep subprime)
300	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	2	Rule Violation	factually_inconsistent	RULE_VIOLATION	STRICT	OBJECT	●●●●●● ●●● 1.00	Subject and predicate match a DB candidate, but the object '2' contradicts the stored value '22.94'. Ev: (300 INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE 22.94)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
303	INTEREST_RATE_CHARGED_AS_ANNUALBORROWING_COST_PERCENTAGE	3	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■■ ■■■ 0.00	No DB candidates found for the subject '303' with the given predicate.
333	INTEREST_RATE_CHARGED_AS_ANNUALBORROWING_COST_PERCENTAGE	4	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■■ ■■■ 0.00	No DB candidates found for the subject '333' with the given predicate.
777	INTEREST_RATE_CHARGED_AS_ANNUALBORROWING_COST_PERCENTAGE	6	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	■■■■■■■■ ■■■ 0.00	No DB candidates found for the subject '777' with the given predicate.
loan_100000_usd	HAS_REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require a certain debt-to-income ratio, collateral, or savings (e.g., 10-20% of the loan amount), but this depends on the lender's policy.	Unverifiable	unverifiable	—	NONE	—	■■■■■■■■ ■■■ 0.00	The predicate is not known in the DB schema and no candidates are available.
300	HAS_DESCRIPTION	Deep subprime (very poor credit score, indicating high risk of default and limited access to credit)	Unverifiable	unverifiable	—	NONE	—	■■■■■■■■ ■■■ 0.00	No DB candidates exist for the subject, predicate, or object.
300	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	1	Rule Violation	factually_inconsistent	RULE_VIOLATION	STRICT	OBJECT	●●●●●● ●●● 1.00	Subject and predicate match the DB candidate, but the object '1' contradicts the stored value '8'. Ev: subject: 300, predicate: APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD, object: 8

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
305	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	2	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	OBJECT	●●●●●■ ■■■ 0.60	Subject exists in DB, but the predicate 'APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD' is not associated with it. Ev: subject: 305, predicate: RISK_TIE_R_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 305, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 22.61
730	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	4	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	OBJECT	●●●●●■ ■■■ 0.60	Subject exists in DB, but the predicate 'APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD' is not associated with it. Ev: subject: 730, predicate: RISK_TIE_R_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: prime; subject: 730, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 7.29

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

No logical inconsistencies detected.

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Multi-Rule Violation	Violation spans multiple knowledge-graph triples simultaneously.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship absent from the knowledge graph entirely.